12/27 Hajime Nakahashi

Toric GIT

ref Cox, Little, Shenck §14.1 ~ §14.3

Mukai "An introduction to Invariants and moduli" §6.1, §6.3

We consider GIT of $\begin{cases} G \subseteq (\mathbb{C}^*)^r : \text{closed subgr} \\ G \curvearrowright \mathbb{C}^r \end{cases}$

which uses χ : character of G.

more precisely Proj quot in direction χ.

$$\mathbb{C}^r /\!/_\chi G := \operatorname{Proj} R_\chi$$

$$\text{where } R_\chi = \bigoplus_{d \geq 0} R^G_{\chi^d}$$

$R^G_{\chi^d}$ = {semi inv. of weight χ^d i.e. $F(g \cdot t) = \chi^d(g) F(t)$}

Aim

⟨1⟩ show GIT $\mathbb{C}^r /\!/_\chi G$ is a semi proj toric var.

⟨2⟩ How to descrive a given toric var in GIT form.

⟨3⟩ what happens as χ varies

Motivation

- When χ moves, Proj quot $X /\!/_\chi G$ undergoes a birat. transformation (A flop / flip is a special case of this)

  ⇝ Our final goal is Toric MMP.

- In general $X /\!/_\chi G \cong (X)^{ss}_\chi / G$ good categorical quot.

  We already know one of example ; Quotient const.

  $X_\Sigma \cong \mathbb{C}^{\Sigma(1)} - Z(\Sigma) /\!/ G$ almost geom. quot.

  in ⟨2⟩ we show $\mathbb{C}^{\Sigma(1)} - Z(\Sigma) = (\mathbb{C}^{\Sigma(1)})^{ss}_\chi$ for some χ

Quick review of GIT

Linearized line bundle

$\mathcal{L}_\chi$: the sheaf of sections of $\mathbb{C}^r \times \mathbb{C}$ with G-action

$s \in \Gamma(\mathbb{C}^r, \mathcal{L}_\chi)$ can be written

$s(p) = (p, F(p)) \in \mathbb{C}^r \times \mathbb{C} \quad (p \in \mathbb{C}^r)$

For $\exists! F \in \mathbb{C}[x_1, \cdots, x_r]$ w/ G-action

$$(g \cdot s)(p) = (p, \chi(g) F(g^{-1} \cdot p))$$

Thus $\Gamma(\mathbb{C}^r, \mathcal{L}_\chi)^G \simeq \{F \in \mathbb{C}[x_1, \cdots, x_r] \mid F(g \cdot p) = \chi(g) F(p)\}$

$= \{\text{semi inv of weight } \chi\}$.

Def

- $p \in \mathbb{C}^r$ is semi-stable $\overset{\text{def}}{\Longleftrightarrow}$ $\exists d > 0$, $s \in \Gamma(\mathbb{C}^r, \mathcal{L}_{\chi^d})^G$ s.t $p \in (\mathbb{C}^r)_s := \{p \in \mathbb{C}^r \mid s(p) \neq 0\}$

  $\Leftrightarrow F(p) \neq 0$.

- $p \in \mathbb{C}^r$ is stable $\overset{\text{def}}{\Longleftrightarrow}$ ____________________

  ____

  and G_p : finite

  $\forall$ G-orbit in $(\mathbb{C}^r)_s$ are closed in $(\mathbb{C}^r)_s$

e.g. $\mathbb{C}^* \underset{\text{usual}}{\curvearrowright} \mathbb{C}^r$, $\chi(t) = t^d$

$$\begin{cases} d > 0 : (\mathbb{C}^r)^{ss}_\chi = (\mathbb{C}^r)^s_\chi = \mathbb{C}^r \setminus \{0\} \\ d = 0 : (\mathbb{C}^r)^{ss}_\chi = \mathbb{C}^r, \ (\mathbb{C}^r)^s_\chi = \phi \\ d < 0 : (\mathbb{C}^r)^{ss}_\chi = (\mathbb{C}^r)^s_\chi = \phi \end{cases}$$

This shows (semi-)stable depends strongly on choice of χ

$$R_\chi = \bigoplus_{d \geq 0} \Gamma(\mathbb{C}^r, L_{\chi^d})^G$$

Lem 14.1.10

$\boxed{R_\chi : \text{f.g } \mathbb{C}\text{-alg}}$

<pf> $\because$ G : reductive

Def

$\boxed{\text{GIT quot } \mathbb{C}^r /\!/_\chi G = \mathrm{Proj}(R_\chi)}$

Prp 14.1.12

(1) $\exists$ proj morph $\mathbb{C}^r /\!/_\chi G \longrightarrow \mathbb{C}^r /\!/ G$ $\leftarrow$ χ = trivial

(2) $\mathbb{C}^r /\!/_\chi G \neq 0 \iff (\mathbb{C}^r)^{ss}_\chi \neq \phi$

(3) $\mathbb{C}^r /\!/_\chi G \cong (\mathbb{C}^r)^{ss}_\chi /\!/ G$.

(4) If $(\mathbb{C}^r)^s_\chi \neq \phi \implies (\mathbb{C}^r)^{ss}_\chi /\!/ G$ almost geom quot.

$\dim \mathbb{C}^r /\!/_\chi G = r - \dim G$

(expected dim)

" in general $\leq$ "

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e.g. $G = \{(t,t,u,u) \mid t,u \in \mathbb{C}^*\}$ and $\chi(t,t,u,u) = t$

F is semi inv $\iff F(tx, ty, uz, uw) = t^d F(x,y,z,w)$

$\iff F$ has deg $(d,0)$ in the grading

x, y have $(1,0)$

z, w have $(0,1)$

then $R_\chi \simeq \mathbb{C}[x,y]$

$\implies \mathbb{C}^4 /\!/_\chi G = \mathrm{Proj}(\mathbb{C}[x,y]) \cong \mathbb{P}^1$ expected dim $= 2$

This is because there are No stable point

In fact $(\mathbb{C}^4)^{ss}_\chi = \mathbb{C}^2 \times (\mathbb{C}^2 \setminus \{0\})$, $(\mathbb{C}^4)^s_\chi = \phi$.

In contrast $\chi(t,t,u,u) = t^a u^b$ $(a,b > 0)$

then $\mathbb{C}^4 /\!/_\chi G = \mathbb{P}^1 \times \mathbb{P}^1$, $(\mathbb{C}^4)^s_\chi \neq \phi$!

(1) <u>Goal</u> GIT $\mathbb{C}^r /\!/_\chi G$ is a semi proj toric var.

- we use the construction from polyhedron.
- Since $G \subseteq (\mathbb{C}^*)^r$ and $\widehat{(\mathbb{C}^*)^r} \simeq \mathbb{Z}^r$ we have

$$0 \longrightarrow M \xrightarrow{\delta} \mathbb{Z}^r \xrightarrow{\gamma} \hat{G} \xrightarrow[\text{Lem14.2.1}]{} 0$$

where $\boxed{M = \ker \gamma}$

$\wr$ dual

$$\mathbb{Z}^r \longrightarrow N \qquad e_i \longmapsto \nu_i$$

then $\delta(m) = (\langle m, \nu_1\rangle, \cdots, \langle m, \nu_r\rangle)$ $\quad m \in M$.

For character $\chi = \chi^a \in \hat{G}$ where $a = (a_1, \cdots, a_r) \in \mathbb{Z}^r$

we define

$$\boxed{P_a = \{ m \in M_{\mathbb{R}} \mid \langle m, \nu_i \rangle \geq -a_i \quad 1 \leq i \leq r \} \subseteq M_{\mathbb{R}}}$$

We want to show $\mathbb{C}^r /\!/_\chi G = X_{P_a}$ with torus

$T_N = N \otimes_{\mathbb{Z}} \mathbb{C}^*$.

In order construct toric var from polyhedron P

we need
- ① its recession cone is strongly convex $\longrightarrow$ ok (Lem 14.2.2)
- ② its vertices are lattice points.
- ③ $\dim P = \dim M_{\mathbb{R}}$.

Counter e.g for ②

$$G = \{ (t, t^{-1}, u) \mid t \in \mathbb{C}^*, u = \pm 1 \}$$

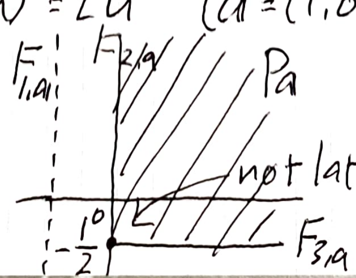
$$0 \longrightarrow \mathbb{Z}^2 \xrightarrow{\delta} \mathbb{Z}^3 \xrightarrow{\gamma} \mathbb{Z} \oplus \mathbb{Z}/2\mathbb{Z}$$

$$(a, b, c) \longmapsto (a-b, c \bmod 2)$$

$\delta(m) = (\langle m, e_1\rangle, \langle m, e_1\rangle, \langle m, 2e_2\rangle)$ $\quad \nu_1 = \nu_2 = e_1$, $\nu_2 = 2e_2$

We choose $\chi(t, t^{-1}, u) = tu$ $\quad (a = (1, 0, 1))$

$$P_a = \begin{cases} \langle m, e_1 \rangle \geq -1 \\ \langle m, e_1 \rangle \geq 0 \\ \langle m, 2e_2 \rangle \geq -1 \end{cases}$$



virtual Faset $F_{i,a} = \{ m \in P_a \mid \langle m, \nu_i \rangle = -a_i \}$

Counter e.g for ③

$G = \{(t,t,u,u) \mid t,u \in \mathbb{C}^*\}$, $\chi(t,t,u,u) = t$ $(a = (1,0,0,0))$

$P_a = \mathrm{Conv}(0, e_1) = [0,1]$

not full dim.

To adress ②; Veronese subring

idea: the vertices are rational, so some integer multiple will be a lattice polyhedral.

$\rightsquigarrow$ translated in algebra.

e.g $G = \{(t,t^{-1},u) \mid u = \pm 1\}$ $\chi(t,t^{-1},u) = tu$

easily to see $(\mathbb{C}^3)^{ss} = \mathbb{C}^* \times \mathbb{C}^2$ ($\because x^2$: semi inv)

$\rightsquigarrow \mathbb{C}^3/\!/_\chi G \simeq (\mathbb{C}^* \times \mathbb{C}^2)/\!/G = \mathrm{Spec}(\mathbb{C}[x^{\pm 1}, y, z]^G)$

$= \mathrm{Spec}(\mathbb{C}[xy, z^2]) \simeq \mathbb{C}^2$

Working directly from $\mathrm{Proj}(R_\chi)$ is harder, because

$R_\chi = \mathbb{C}[xy, z^2] \oplus xz\mathbb{C}[xy, z^2] \oplus x^2\mathbb{C}[xy, z^2] \oplus x^3z\mathbb{C}[xy, z^2] \oplus \cdots$

but we take Veronese subring

$R_\chi^{[2]} = \mathbb{C}[xy, z^2] \oplus x^2\mathbb{C}[xy, z^2] \oplus x^4\mathbb{C}[xy, z^2] \oplus \cdots$

$= \mathbb{C}[xy, z^2][x^2]$

$\mathbb{C}^3/\!/_\chi G = \mathrm{Proj}(R_\chi) \simeq \mathrm{Proj}(R_\chi^{[2]}) = \mathrm{Spec}(\mathbb{C}[xy, z^2]) \simeq \mathbb{C}^2$

Point!

To adress ③; We construct the normal fan of P as a generalized fan (not required "cone is strongly convex")

See §6.2

$\Rightarrow C(P_a) \subseteq M_{\mathbb{R}} \times \mathbb{R}$ whose slice at height $\lambda > 0$ (resp. $\lambda = 0$) is λP_a (resp. the reccession cone of P_a)

$\rightsquigarrow S_{P_a} := \mathbb{C}[C(P_a) \cap (M \times \mathbb{Z})]$

$\rightsquigarrow X_{P_a} \simeq \mathrm{Proj}(S_{P_a})$: semiproj (Prop 14.2.12).

Thm 14.2.13

(1) $R_\chi \simeq \mathbb{C}[C(P_a) \cap (M \times \mathbb{Z})]$

(2) $\mathbb{C}^r /\!\!/_\chi G = \mathrm{Proj}(R_\chi)$ is the toric var of P_a

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⟨2⟩ toric var $\longrightarrow$ GIT

$X_\Sigma = (\mathbb{C}^{\Sigma(1)} - Z(\Sigma))/G$ where $G = \mathrm{Hom}_{\mathbb{Z}}(Cl(X_\Sigma), \mathbb{C}^*)$

$\hat{G} \simeq Cl(X_\Sigma)$

evaluation of β $\longleftarrow\!\shortmid$ β

$a = (a_\rho) \longleftarrow\!\shortmid [D] = [\sum a_\rho D_\rho]$

Prop 14.1.9

If $\chi \in \hat{G}$ comes from an ample divisor class.

then $\begin{cases} (\mathbb{C}^{\Sigma(1)})^{ss}_\chi = \mathbb{C}^{\Sigma(1)} - Z(\Sigma) \\ (\mathbb{C}^{\Sigma(1)})^{s}_\chi = \mathbb{C}^{\Sigma(1)} - Z(\Sigma') \end{cases}$ $\Sigma' \subseteq \Sigma$ sub fan of all simplicial cones.

$P_a = P_D$: generalized fan ass. to Cartier D.

when $\begin{cases} \cdot D \text{ is ample} & \mathbb{C}^r /\!\!/_\chi G \simeq X_\Sigma \\ \cdot D \text{ is nef} & \mathbb{C}^r /\!\!/_\chi G \simeq X_{P_D} \end{cases}$ (see Eg 14.2.14)

③ what happens to the GIT quot $\mathbb{C}^r /\!/_\chi G$ as we vary χ $\boxed{7}$

<u>Key</u> Gale duality

$$0 \to V \xrightarrow{\delta} \mathbb{R}^r \xrightarrow{\gamma} W \to 0 \quad \text{vect. sp}$$

$$W = \langle \beta \rangle = \langle \beta_1, \cdots, \beta_r \rangle$$

dual

$$0 \to W^* \xrightarrow{\gamma^*} \mathbb{R}^r \xrightarrow{\delta^*} V^* \to 0$$

$$\nu_i = \delta^*(e_i)$$

$$\Rightarrow \delta(v) = (\langle v, \nu_1 \rangle \cdots \langle v, \nu_r \rangle)$$

<u>Property 1</u> $I \subseteq \{1, \cdots, r\}$, $J = \{1, \cdots, r\} \setminus I$

β_i $(i \in I)$ form a basis of $W \Leftrightarrow \nu_j$ $(j \in J)$ form a basis of V^*

$C_\beta := \mathrm{Cone}(\beta) \subseteq W$.

$C_\nu := \mathrm{Cone}(\nu) \subseteq V^*$

<u>Property 2</u> (Gale duality)

- If $C_\beta = W$, then C_ν is strongly convex in V^*. Conversely if ν consists of nonzero vectors and C_ν is strongly convex $\Rightarrow C_\beta = W$.
- If $C_\nu = V^*$, then ______

we apply this to

$$0 \to M \to \mathbb{Z}^{\Sigma(1)} \to \hat{G} \to 0$$

$\otimes_{\mathbb{Z}} \mathbb{R}$

$$0 \to M_{\mathbb{R}} \to \mathbb{R}^{\Sigma(1)} \to \hat{G}_{\mathbb{R}} \to 0.$$

C_β controls "When (semi)stable point exists"

Prop 14.3.5 TFAE

(a) $\mathbb{C}^r /\!\!/_\chi G \neq \emptyset$

(b) $(\mathbb{C}^r)^{ss}_\chi \neq \emptyset$

(c) $\chi \otimes 1 \in C_\beta$

Prop 14.3.6 TFAE

(a) $(\mathbb{C}^r)^{s}_\chi \neq \emptyset$

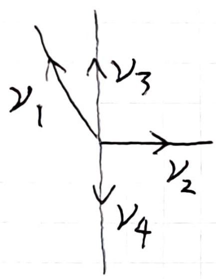
(b) $\chi \otimes 1 \in C_\beta^\circ$

(c) $F_{z,a}$ is a proper face of P_a

$\Rightarrow \dim \mathbb{C}^r /\!\!/_\chi G = r - \dim G$.

e.g. $G = \{(t, t^{-1}u, t, u) \mid t, u \in \mathbb{C}^*\} \rightsquigarrow Cl(\mathcal{H}_r)$

$$0 \longrightarrow \mathbb{Z}^2 \xrightarrow{B} \mathbb{Z}^4 \xrightarrow{A} \mathbb{Z}^2 \longrightarrow 0$$

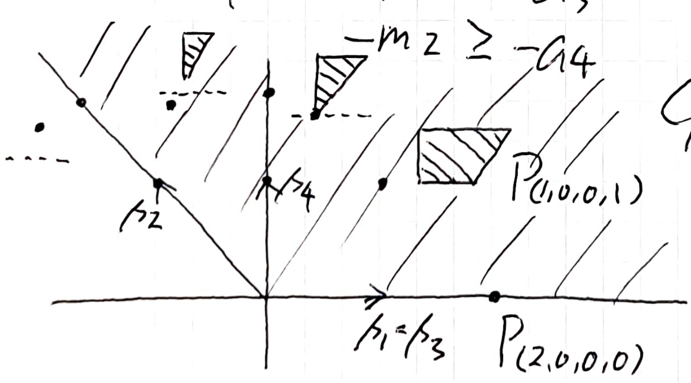


$$\begin{array}{c} \nu_1 \\ \nu_2 \\ \nu_3 \\ \nu_4 \end{array} \begin{pmatrix} -1 & r \\ 0 & 1 \\ 1 & 0 \\ 0 & -1 \end{pmatrix} \qquad \begin{array}{c} \begin{pmatrix} 1 & -r & 1 & 0 \\ 0 & 1 & 0 & 1 \end{pmatrix} \\ \beta_1 \quad \beta_2 \quad \beta_3 \quad \beta_4 \end{array} \qquad AB = 0$$

For $\underline{r = 1}$

$$P_a = \begin{cases} -m_1 + m_2 \geq -a_1 \\ m_2 \geq -a_2 \\ m_3 \geq -a_3 \\ -m_2 \geq -a_4 \end{cases}$$

$(a_1, a_2, a_3, a_4) \longmapsto \sum a_i \beta_i$



$\cdots\cdots$: vertual faset.

 $\leftarrow$ not generic

i.e $\forall$ virtual faset is $\begin{cases} \text{empty} \\ \text{genuine faset.} \end{cases}$

We can expect the existence of walls and chambers $\longrightarrow$ Next speaker...

Proj morph $\mathbb{C}^r/\!/_\chi G \longrightarrow \mathbb{C}^r/\!/G$

extreme case

$\boxed{\mathbb{C}^r/\!/G \text{ is very small}}$

Prop 14.3.10 TFAE

(a) $\mathbb{C}^r/\!/_\chi G$ is proj for all $\chi \otimes 1 \in C_\beta$.

(b) ——— some ———

(c) $\mathbb{C}[x_1, \dots, x_r]^G = \mathbb{C}$

(d) β consists of nonzero vectors and C_β is strongly convex. ($\Leftrightarrow$ $C_\nu = V^*$)

$\boxed{\mathbb{C}^r/\!/G \text{ is very large}}$

Prop 14.3.11 TFAE

(a) $\dim \mathbb{C}^r/\!/G = r - \dim G$.

(b) C_ν is strongly convex ($\Leftrightarrow$ $C_\beta = W$)

(c) $\mathbb{C}^r/\!/_\chi G \longrightarrow \mathbb{C}^r/\!/G$ birat. for all $\chi \otimes 1 \in C_\beta$.